

SANGBUNHAK

Sangbun Vector Computation and Graph Construction

Authoritative Text · Revised Edition

Text-only edition (graphs omitted)

Overview

Sangbunhak is a geometric and vector methodology that interprets, step by step, the geometric structure defined at a single point and the vector relations generated around that point.

Tangent, normal, projection, rotation and vector sum — geometric processes that all start from a point — are linked to one another, and the formation of a single vector structure out of them is explored through an intuitive, figure-based reading.

The Sangbun vector is a double-structured vector combining a straight-line geometric component generated by the tangent and projection structure with a rotational geometric component generated by the circle and rotation structure.

Premises

- The vector is computed at a single point on a line or a curve.
- Every construction below assumes a non-degenerate general position.
- Special cases in which an intersection is undefined are defined by separate rules.

Z1 Computation

Basic setup

Slope : a

Origin : $O(0, 0)$

Tangent line : $y = ax + b$

Point : $A(x, ax + b)$

Basic projections and intersections

A1 — foot of the perpendicular dropped from O to the tangent line

A2 — foot of the perpendicular dropped from O to the normal line at A

A3a — intersection of the normal line at A with the x-axis

$A3a' = (x\text{-coordinate of } A3a, y\text{-coordinate of } A1)$

A3b — intersection of the normal line at A with the y-axis

$A3b' = (x\text{-coordinate of } A1, y\text{-coordinate of } A3b)$

A3c — intersection of the vertical line through A3a with the tangent line

A3d — intersection of the horizontal line through A3b with the tangent line

Feet of perpendiculars taken from the origin

A4a — intersection of line OA3c with line AA3a'

A4b — intersection of line OA3d with line AA3b'

Construction of A6 (vector sum)

A5a — intersection of line (1), the line through A2 perpendicular to A1A2, with the extension of A1A4a

A5b — intersection of line (1) with the extension of A1A4b

$A1A6 = A1A5a + A1A5b$

Circle (1) and the principal points

Circle (1) — center A1, radius $|A1A|$

A7 — the point where the line running from O in the direction of A1 meets Circle (1) after passing through A1

A8 — the intersection, other than A, of Circle (1) with the line through A perpendicular to the vector OA

A9 — of the two points lying at distance $|A7A8|/2$ perpendicularly from the midpoint of segment A7A8, the one inside Circle (1)

A9a — of the same two points, the one outside Circle (1)

Branch criterion value

$$@ = |OA1| / |AA1|$$

Unified Definition of A11 and the Necessity of Line A10'A11

1) Definition of A11

A, A10 and A10' form a triangle with its right angle at A10. The foot of the perpendicular dropped from A10 to the hypotenuse AA10' is A9 when $@ < 1$ and A7 when $@ > 1$.

A11 is the point obtained by rotating A10 through 90 degrees about A. The sense of rotation is fixed by the turn that carries the direction from A toward O onto the direction from A toward A10: the same sense when $@ < 1$ and when $@ = 1$, the opposite sense when $@ > 1$.

$$|AA10| = |AA11|$$

2) How R differs

R is an auxiliary point obtained by rotating the same A10 through 90 degrees in the opposite sense about A10'. A11 and R rotate the same point but about different centers. Accordingly A11 preserves the length measured from A, while R preserves the length measured from A10'.

3) Definition of Q

Q is the intersection of the two lines AR and the perpendicular A10A9 (A10A7 when $@ > 1$). Q is determined by A, A10 and A10' alone and does not use A11.

4) Necessity of line A10'A11

Line A10'A11 passes through this same Q. The three lines therefore meet at one point, and no point of the authoritative text other than Q lies on this line. Hence joining A10' to A11 is not an arbitrary auxiliary line but a necessary connection that exhibits A11 as lying on the concurrency structure.

5) Scope of application

R and Q are not points of the authoritative text but auxiliary points used to confirm the position of A11, and they are not shown in the graph.

When $@ = 1$, the points from A7 through A10' coincide in a single point and the right triangle degenerates; this auxiliary structure is therefore not applied, and only the rotation rule of 1) is applied.

Case in which the sense of rotation is undefined — if the direction from A toward O and the direction from A toward A10 coincide or are exactly opposite, the sense of the turn is not determined. In that case A11 is defined by a separate rule.

Branch 1 : $@ < 1$

Circle (2) — center A8, radius $|A8A9|$

A10 — the intersection, other than A9, of Circle (2) with the line through A9 perpendicular to AA9

A10' — intersection of the line through A10 perpendicular to AA10 with the extension of AA9

A11 — as given by the unified definition

Branch 2 : $@ > 1$

Circle (2) — center A9, radius $|A9A7|$

A10 — the intersection, other than A7, of Circle (2) with the line through A7 perpendicular to AA7

A10' — intersection of the line through A10 perpendicular to AA10 with the extension of AA7

A11 — as given by the unified definition

Branch 3 : $@ = 1$

Rule for selecting A (inequality conditions)

Once a and b are given, two points on the tangent line satisfy $@ = 1$, one on each side of A1. When drawing a graph or computing, one of the following four conditions is specified so that A(X, Y) is fixed uniquely. The coordinates of A1 are A1(A1X, A1Y).

1) $X > A1X, Y < A1Y$

2) $X < A1X, Y > A1Y$

3) $X > A1X, Y > A1Y$

4) $X < A1X, Y < A1Y$

- The two points lie one on each side of A1 along the tangent line. Hence only 3) or 4) can hold when the slope a is positive, and only 1) or 2) when it is negative.

- If the tangent line is horizontal, the two points share the y-coordinate of A1, so the choice is made by the x condition alone. The cases in which the reference line is vertical or horizontal are treated separately under the Z2 computation.
- The same selection rule applies when finding a point with $@ = 1$ on a curve (a function).

$$A7 = A8 = A9 = A9a = A10 = A10'$$

A11 — as given by the unified definition. Since $A10 = A7$,
 $|AA10| = |AA7| = |AA11|$

Final point A12

$$AA12 = AA6 + AA11$$

Special cases

1) The case $y = ax$, $A(x, ax)$ ($b = 0$)

Since $A1 = O$, the point A7 is undefined. Taking the direction that bisects the two limits as b approaches 0 from the right and from the left ($= AO$), A11 is defined to be O.

$A12 = A$ (complete self-closure structure)

- This is an extension by convention; it is not a statement that the construction is continuous at $b = 0$.

2) The case $A = A1$ (OA perpendicular to the tangent line, $|AA1| = 0$)

@ is undefined, and the radius of Circle (1) becomes 0, so no point from A7 onward can be generated.

A11 cannot be generated $\rightarrow AA12 = AA6 \rightarrow A12 = A6$

3) The case in which the direction from A toward O is parallel to the direction toward A10

If the two directions coincide or are exactly opposite, the sense of the turn producing A11 is not determined. It is defined by a separate rule.

Z1 Graph Construction

Basic setup

Mark the origin $O = (0, 0)$

Axis ratio 1 : 1

Grid spacing 1

Draw and label the x-axis and the y-axis

Reference elements

Draw the tangent line $y = ax + b$

Mark the point $A(x, ax + b)$

Mark the point A1 (foot of the perpendicular from O to the tangent line)

Circles

Circle (1) — center A1, radius $|A1A|$

Circle (2), $@ < 1$ — center A8, radius $|A8A9|$

Circle (2), $@ > 1$ — center A9, radius $|A9A7|$

Circle (2), $@ = 1$ — undefined

Point label rules

Points to be labeled

O, A, A1, A2, A4a, A4b, A5a, A5b, A6, A7, A8, A9, A9a, A10, A10', A11, A12

Points not to be labeled

A3a, A3a', A3b, A3b', A3c, A3d

Coordinate notation

Coordinates are shown for A, A1, A10' and A12 only

Five decimal places

Show a, b, x and @ at one side of the graph (when @ = 1, show the selected inequality condition number in place of x)

Standard connecting lines (always drawn)

OA, OA1, OA2, A1A2, A1A6, A1A7, AA2, AA6, AA9, AA10, AA11, AA12, A1A5a, A1A5b, A2A4a, A2A4b, A2A5a, A2A5b, A5aA6, A5bA6, A6A12, A11A12, A7A9, A7A9a, A8A9, A8A9a, A10A10', A10'A11

Additional connecting lines by @

@ < 1 — AA8, A9A10, A9A10', OA8

@ > 1 — AA7, A7A10, A7A10', OA7

@ = 1 — since $A7 = A8 = A9 = A9a = A10 = A10'$, label A10' only; the connecting line is AA10'

Graph notation for special cases

A = A1 — write on the graph: “no @ value ($A = A1$, $|AA1| = 0$)” and “A11 cannot be generated $\rightarrow AA12 = AA6 \rightarrow A12 = A6$ ”, and omit Circle (1), Circle (2) and every point and connecting line from A7 onward.

Z2 Computation

Definition of Z2

The case in which the reference line is vertical or horizontal.

$$x = k$$

$$y = c$$

The point A takes one of the following four forms

$$\text{for } x = k, A = (k, k_1)$$

$$\text{for } y = c, A = (c_1, c)$$

$$\text{for } x = k, A = (k, k)$$

$$\text{for } y = c, A = (c, c)$$

$$\text{Origin } O = (0, 0)$$

A1 (always the same)

Foot of the perpendicular from O to the reference line

$$x = k \rightarrow A1 = (k, 0)$$

$$y = c \rightarrow A1 = (0, c)$$

A2 (axis projection of A)

$$x = k, A = (k, k_1) \rightarrow A2 = (0, k_1)$$

$$x = k, A = (k, k) \rightarrow A2 = (0, k)$$

$$y = c, A = (c_1, c) \rightarrow A2 = (c_1, 0)$$

$$y = c, A = (c, c) \rightarrow A2 = (c, 0)$$

Circle (1) and A6

Circle (1) — center A1, radius $|A1A|$

A6 — intersection of the line through A2 perpendicular to A1A2 with the reference line

Here the reference line is the vertical line when $x = k$ and the horizontal line when $y = c$

- The definition of @ is the same as in Z1, and the stages after $@ < 1$, $@ > 1$ and $@ = 1$ follow the same branch rules as in Z1.

Selecting the condition when $@ = 1$ (Z2)

For $x = k$ we have $A = (k, k_1)$ and $A_1 = (k, 0)$, so the two points share the same x-coordinate and $@ = 1$ means that the absolute value of k_1 equals the absolute value of k .

The choice is between $Y > A_1Y$ ($k_1 > 0$) and $Y < A_1Y$ ($k_1 < 0$).

For $y = c$ we have $A = (c_1, c)$ and $A_1 = (0, c)$, so the two points share the same y-coordinate and $@ = 1$ means that the absolute value of c_1 equals the absolute value of c .

The choice is between $X > A_1X$ ($c_1 > 0$) and $X < A_1X$ ($c_1 < 0$).

Thus in Z2 the four conditions of Z1 reduce to two.

Computing A7 through A12 (core rule)

In Z2 the computation from A7 onward is carried out in exactly the same way as in Z1.

$A_7, A_8, A_9, A_{9a}, A_{10}, A_{10'}, A_{11}, A_{12}$ = vector sum

- Only the reference line differs.
- The rules for rotation, Circle (2) and the vector sum are entirely those of Z1.
- A_{11} likewise follows the unified definition without change.

A special consequence in Z2

The line $y = ax$ has a defined slope and is therefore an ordinary line belonging to Z1. The resulting self-closure ($A_{12} = A$ when $b = 0$) is defined under Z1 special case 1) and is not treated separately in Z2.

Z2 Graph Construction

Basic display

x-axis, y-axis, origin $O(0, 0)$

Axis ratio 1 : 1

Grid spacing 1

Reference line and points

Reference line : $x = k$ or $y = c$

Points : A, A1, A2, A6, A7, A8, A9, A9a, A10, A10', A11, A12

Circles

Circle (1) — center A1, radius $|A1A|$

Circle (2), $@ < 1$ — center A8, radius $|A8A9|$

Circle (2), $@ > 1$ — center A9, radius $|A9A7|$

Circle (2), $@ = 1$ — omitted

Standard connecting lines

OA, A1A2, AA2, A2A6, AA6, A6A12, AA9, AA10, AA11, AA12, A7A9, A7A9a, A8A9, A8A9a, A11A10', A10A10'

Additional connecting lines by @

$@ < 1$ — AA8, A9A10, A9A10', OA8

$@ > 1$ — AA7, A7A10, A7A10', OA7

$@ = 1$ — since $A7 = A8 = A9 = A9a = A10 = A10'$, label A10' only; the connecting line is AA10'

Coordinates and notation

Coordinates shown for A, A1, A10' and A12 only

Five decimal places

At one side of the graph: k or c, the value of @, the coordinates of A12, $|AA12|$, and the angle of AA12 (when $@ = 1$, also show the selected condition number)